Assessment Tool - 2

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Question 1

Design a Machine Learning System for Health Analysis

A machine learning system can be developed to predict diseases such as **heart disease** or **diabetes** using patient information like **age, blood pressure, glucose level, cholesterol, BMI, and lifestyle factors (smoking, exercise, diet, etc.)**.

Suitable Algorithms

1. **Decision Tree** – Easy to understand and can handle both numerical and categorical data.
2. **Random Forest** – Provides higher accuracy by combining multiple decision trees and reduces overfitting.
3. **Logistic Regression** – Suitable for binary classification problems such as predicting whether a disease is present or not.

Justification:
Among these, **Random Forest** is the preferred algorithm because it provides high accuracy, handles missing values effectively, and is less prone to overfitting compared to a single decision tree.

Data Preprocessing

- Collect patient health records from reliable sources.
- Handle missing values by removing or replacing them.
- Remove duplicate and inconsistent records.
- Convert categorical values into numerical values.

- Normalize or standardize numerical features to improve model performance.

Feature Selection

Select the most important features such as:

- Age
- Blood Pressure
- Glucose Level
- BMI
- Cholesterol
- Smoking Habit
- Physical Activity
- Family History

Feature selection helps reduce irrelevant data and improves prediction accuracy.

Model Training

- Split the dataset into **80% training data** and **20% testing data**.
- Train the selected machine learning algorithm using the training dataset.
- Test the model using the testing dataset.
- Evaluate performance using **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**.
- Fine-tune the model to improve prediction performance.

Conclusion

A machine learning-based health analysis system can accurately predict diseases by using proper preprocessing, feature selection, and an efficient algorithm such as Random Forest, helping doctors make faster and more reliable decisions.

Question 2

Apply the Find-S Algorithm

Step 1: Training Dataset

1. (Employed, High, Good, Clean) → **Yes**
2. (Employed, Medium, Good, Clean) → **Yes**

3. (Unemployed, Low, Poor, Bad) → **No** *(Ignored because Find-S uses only positive examples)*
4. (Employed, Medium, Excellent, Clean) → **Yes**

Step 2: Initial Hypothesis

$$H_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

Step 3: Process Positive Example 1

Example:

(Employed, High, Good, Clean)

$$H_1 = \langle Employed,\ High,\ Good,\ Clean \rangle$$

Step 4: Process Positive Example 2

Example:

(Employed, Medium, Good, Clean)

Compare with H_1:

- Employment → Same → **Employed**
- Salary → High ≠ Medium → **?**
- Credit Score → Same → **Good**
- Loan History → Same → **Clean**

$$H_2 = \langle Employed,\ ?,\ Good,\ Clean \rangle$$

Step 5: Ignore Negative Example

Example 3 is negative (**No**), so it is ignored by the Find-S algorithm.

$$H_2 = \langle Employed,\ ?,\ Good,\ Clean \rangle$$

Step 6: Process Positive Example 4

Example:

(Employed, Medium, Excellent, Clean)

Compare with H_2:

- Employment → Same → **Employed**
- Salary → ? remains **?**
- Credit Score → Good ≠ Excellent → **?**
- Loan History → Same → **Clean**

$$H_3 = \langle Employed,\ ?,\ ?,\ Clean \rangle$$

Final Hypothesis

$$H = \langle Employed,\ ?,\ ?,\ Clean \rangle$$

Conclusion

The Find-S algorithm learns that a loan is likely to be approved if the applicant is **Employed** and has a **Clean Loan History**, while **Salary** and **Credit Score** can take any value (?) because they vary among the positive examples.